



Section A: Structured Answers

QUESTION 1

(a) Fill in the net change in average charge for mutant hemoglobin C and $M_{\text{Milwaukee}}$ in **Figure 1.1**. [1]
+2, -1

(b) A sample of the hemoglobin of a patient known to be suffering from a genetic defect in the β chain is subjected to electrophoresis at pH 8 along with a sample of normal hemoglobin A. The patient's hemoglobin migrates towards the anode faster than hemoglobin A. Which of the β -chain mutants listed in **Figure 1.1** could the patient's hemoglobin be? Briefly explain your answer. [2]

$M_{\text{Milwaukee}}$. That the patient's hemoglobin migrates more rapidly towards the positive anode indicates that it has a greater negative charge than hemoglobin A and since only the substitution of Glu for Val in $M_{\text{Milwaukee}}$ leads to a more negative protein, $M_{\text{Milwaukee}}$ must be the β -chain mutant.

(c) Given that position 67 lies in the cleft of the β chain where the heme functional group is wedged, deduce if the substitution of Glu for Val in $M_{\text{Milwaukee}}$ leads to any effect on the stability of the hemoglobin molecule. Justify your answer. [2]

Substitution will destabilise the protein / cause overall stability of the protein to decrease. Replacement of hydrophobic Val with negatively-charged Glu results in burying of a lone charge in hydrophobic cleft / interior of the β chain; which is thermodynamically / energetically unfavourable due to lack of bonds formed between the charged side chain and other mostly hydrophobic groups in the cleft.

(d) The relative molecular mass of human hemoglobin is 68000 and the molecules are contained in red blood cells. The relative molecular mass of the hemoglobin of a species of worm is 2.85 million and the molecules are found in the plasma. Suggest one advantage for the relatively huge size of hemoglobin when it is in the plasma. [1]

Hemoglobin molecules do not escape from the blood / do not pass through walls of blood vessels / are not excreted / do not have osmotic effect.

(e) Hemoglobin is an intracellular protein consisting of α and β subunits predominantly made up of lengths of helical regions. Give an example of an extracellular protein that has a helical conformation, and discuss the significance of the protein's helical nature to its function. [4]

α -keratin. Pairs of α helices which are wound around each other to form a dimer with a coiled-coil conformation which confers high tensile strength and stability particularly with assemblage of dimers into protofilaments and protofibrils.

Ref to three / seven α helices coiled around one another held by disulphide cross linkages to form strong and tough fibres.

Ref to the extensive hydrogen bonding involving all peptide NH and CO groups in the α helical subunit, which confers strength / stability and insolubility in water.

Ref to structural function of the protein.

OR

Collagen. Three helical α chains wound around one another to form a coiled-coil triple helix maintained by interchain hydrogen bonds which confers high tensile strength rigidity / inflexibility and stability particularly with packing of tropocollagen into collagen fibres.

Ref to the extensive interchain hydrogen bonding involving all peptide NH and CO groups and covalent interchain cross-links in the triple helical subunit, which confers strength / stability and insolubility in water.

Ref to structural function of the protein.

[Total: 10 marks]

QUESTION 2

(a) Calculate the specific activity, fold-purification and the percentage yield for each step as compared to the initial starting material and fill in the values in **Table 2.1** on the insert provided. [3]

- **Specific Activity: Divide total enzyme activity by total protein**
- **Fold Purification: Divide sp. act. by 8.2 (initial sp. act.)**
- **% Yield: Divide total enzyme activity by starting total activity x 100**



Steps	TOTAL RECOVERY		SPECIFIC ACTIVITY	FOLD PURIFIED	% YIELD
	PROTEIN (mg)	ENZYME (units)			
Student X					
Initial Starting Material	511	4200	8.2	-	-
40 % Ammonium sulphate	181	3991	22	2.7	95
DEAE (Anion exchange chromatography) flow through, pH 7.0	67.6	3787	56	6.8	90
Isoelectric Focusing (pH 9.7)	3.8	2949	776	95	70
Student Y					
Initial Starting Material	714	5880	8.2	-	-
40 % Ammonium sulphate	253	5584	22	2.7	95
DEAE flow through, pH 9.0	5.1	4800	941	115	82
Student Z					
Initial Starting Material	410	3360	8.2	-	-
40 % Ammonium sulphate	146	3192	22	2.7	95
CM, pH 7.0 NaCl elution	1.9	2912	1533	186	86

- (b) State which student has the best purification scheme and give a reason for your answer. [2]
- **Student Z has the better purification scheme with less steps, better yield and better fold purification.**
 - **Other factors to consider would be whether pH 9 or the NaCl would be detrimental for the subsequent use of the protein.**
- (c) Based on Student X's purification scheme, account for Student Z's decision to use cation exchange chromatography. [1]
- **Isoelectric point of 9.7**
 - **cation exchange purification usually performed below the isoelectric point**
 - **failure to bind to an anion exchange column suggested that a cation exchanger might bind the protein**
- (d) State the approximate size(s) of the protein(s) which is/are likely to be responsible for the enzyme activity. Give a reason for your answer. [2]
- **2 bands of approximately 100 kDa and 50 kDa that are enriched by both gel filtration and isoelectric focusing as compared to the original starting material**
 - **none of the other bands are enriched by both methods**
- (e) Describe the technique which would enable Student X to elucidate real-time interaction of the enzyme with other proteins and the kinetics of these interactions. [2]
- **Surface plasmon resonance (SPR)**
 - **measurable change in the resonance angle due to interaction of flow through proteins with the immobilized bait protein through the biosensor**
 - **a binding constant (K) is calculated by dividing k_{off} by k_{on} .**

QUESTION 3

- (a) Briefly describe the steps highlighted in Box P. [2]
- **Isoelectric focussing / 1st dimension separation by pI / pH**
 - **SDS-PAGE / 2nd dimension separation by molecular weight**
 - **Staining of gel with Coomassie Blue / silver stain**
 - **Digital image of 2D-gel captured / comparison of spot intensities between gels using software**
- (b) Outline the subsequent step(s) not shown in Fig. 3.1, and explain how the results of this experiment may help researchers understand the effects of TGF- β on hMSCs. [4]
- **Core out spots of interest**
 - **Tryptic digest of proteins in these spots**
 - **Subject tryptic digest to MS / (named e.g. MALDI-TOF) MS**
 - **generating peptide mass fingerprint(s) (PMF)**
 - **ptn identification by searching PMFs against database**
 - **(Relative) quantitative comparison of proteins expressed**
 - **Identify which protein upregulated/ downregulated/ differentially expressed in presence of TGF- β**



- Correlate proteome changes to changes in hMSC characteristics
- Deduce the ptns involved in determining MSC characteristics

(c) State and explain a drawback of the experimental methodology shown in Fig.3.1. [2]

- **different run conditions for treated vs non-treated samples**
- **may introduce inter-gel variations**
- **ref inaccuracies in relative quantitation between gels**
- **membrane ptns under/not represented**
- **other logical points**

(d) Briefly describe another methodology that allows the simultaneous analysis of thousands of proteins. [2]

- **Protein microarray**
 - **Analyse/detect protein binding partners of various target proteins**
 - **Immobilised on glass chip/microarray**
- **Shotgun MS-based approach to identifying proteins**
 - **Tryptic digests of complex protein sample e.g. total cell lysate**
 - **Subject to LC-MS**
 - **Generate PMFs for search against PMF database**

QUESTION 4

(a)(i) Using the Lineweaver-Burk plot, determine the apparent k_m and V_{max} values in the presence and absence of *threo*-sphingosine. Present your answer on the graph paper provided. [3]

- **Without *threo*-sphingosine: $k_m = 7.2 \mu M$ $V_{max} = 123.5 \text{ mgmin}^{-1}$**
- **With *threo*-sphingosine : $k_m = 25.8 \mu M$ $V_{max} = 95.2 \text{ mgmin}^{-1}$**
- **$1/K_m$ and $1/V_{max}$ clearly indicated on the graph ;**

(ii) Account for the difference in V_{max} values in the absence and presence of *threo*-sphingosine. [3]

- **V_{max} in the presence of *threo*-sphingosine is lower than in its absence**
- **Because *threo*-sphingosine is a competitive inhibitor**
- **It competes with sphingosine for the active site of the enzyme**
- **Lowering the no. of ES complexes that can be formed per unit time**
- **The same V_{max} can be achieved in the presence and absence of *threo*-sphingosine**
- **But substrate concentration used in the reaction is not high enough/ too low**

(b) Using the data provided, estimate k_i , showing all relevant working. [2]

- **$\alpha k_m = 25.8$, $\alpha = 3.58$, $\alpha = 1 + [I]/k_i$**
- **$k_i = 3.9$**

(c) Using the values obtained in (a)(i) and (B), comment on the effectiveness of *threo*-sphingosine as an inhibitor. [2]

- ***threo*-sphingosine is effective as an inhibitor**
- **k_i is smaller than k_m**
- **showing that enzyme has higher affinity for the *threo*-sphingosine than sphingosine**
- **Formation of enzyme-inhibitor complex is preferred over to enzyme-substrate complex**

QUESTION 5

(a) Briefly explain why acetylcholine receptors are necessarily transmembrane proteins. [1]

- **Acetylcholine is a relatively large charged molecule that is unable to pass through the cell membrane.**

(b)(i) Account for the fact that both nAChRs and mAChRs can bind to acetylcholine even though their primary sequences differ greatly. [2]

Folding of amino acid sequences into secondary, tertiary and quaternary structures via formation of H-bonds, disulfide bonds, ionic interactions and hydrophobic interactions between the R groups of structural amino acids resulting in the 3D configuration of both types of AChRs possessing a ligand binding site of almost identical conformation/charge distribution and thus able to bind to acetylcholine

(ii) State one structural feature of an mAChR that you would not expect to find in an nAChR. [1]

G protein binding site;;

(c)(i) With reference to Fig 4.2, explain why there is a large predominance of hydrophobic amino acids in the sequences of helices M1 and M3. [2]

- **Helices M1 and M3 of each subunit partially face the exterior of the channel/ the phospholipid bilayer**
- **hydrophobic interactions with the fatty acid tails of the bilayer allow for the maintenance of the relative position of the AChR in the membrane**



- Helices M1 and M3 of each subunit also partially face helices M3 and M1 of the adjacent subunits respectively
 - the predominance of hydrophobic aa could allow for hydrophobic interactions to be formed between adjacent subunits holding the quaternary level of the protein structure
- (ii) For all five subunits, helix M2 is positioned such that it faces the channel pore. Describe and explain one difference that you would expect to see between the amino acid sequences of helix M2 and M3. [2]
- **M2 would have a larger predominance of hydrophilic amino acids which would occur at every 4th residue in comparison to the relatively larger predominance of hydrophobic aa in M3**
 - **these hydrophilic aa would face the core of the channel thus allowing for a hydrophilic channel environment that allows the passage of the charged molecule acetylcholine whereas helix M3 does not require hydrophilic aa as it does not interact with acetylcholine**
- (d) Suggest two reasons why leucine is a suitable amino acid for use in the “gate”. [2]
- **Hydrophobic will repel acetylcholine when gate is closed (ie. leucine projects into the interior of the channel)**
 - **Relatively large side chain an effective blockage for the entire channel pore**

Section B: Structured Answers

QUESTION 6

- (a)(i) Explain why the immunoglobulin molecule is referred to as a heterotetramer. [2]
- **Presence of four subunits with two different non-identical subunits, the heavy and light chains.**
- (ii) Outline the importance of the disulphide bridges to the structure of the immunoglobulin molecule. [1]
- **Stabilise each discrete domain**
 - **Hold the two heavy chains together**
 - **Link each of the two light chains to a heavy chain**
 - **Stabilise the quaternary structure / whole tetramer**
- (b) Given that the epitope is protein in nature, describe the bonding interactions that allow for the formation of the antigen-antibody complex shown in **Figure 6.2**. [2]
- Bonding interactions occur between the amino acid side chains / R groups of the antigen epitope and immunoglobulin / heavy or light chains and include ionic interactions between oppositely charged R groups, hydrogen bonds between polar uncharged R groups and hydrophobic interactions between nonpolar residues;**
- (c) With reference to **Figures 6.2 and 6.3**, relate the relative variability of the amino acid residues at the antigen binding site to their significance in antibody function. [3]
- Amino acid residues of the H and L chains at the 29th to 33rd, 50th to 56th and 94th to 99th positions at the antigen binding site exhibit the highest relative variability in sequence which provides the diversity in the 3-dimensional conformation / shape / tertiary structure of the antibody binding site that confer the multitude of specificities; of antibodies to different antigen epitopes.**
- (d) (i) Determine if the IgM molecule is a closed or open structure, and if the interactions between the five units are isologous or heterologous. [1]
- **Closed structure**
 - **Isologous**
- (ii) Suggest why IgM is a more efficient antibody than the one in **Figure 6.1**. [1]
- Due to its pentameric form, it possesses multiple valence/ more than two antigen binding sites/ it can form interactions between epitopes/ antigens distant from one another, thus making it an effective agglutinating antibody.**

QUESTION 7

- (a) Explain the importance of mixing the samples in a 1:1 ratio after labeling prior to subsequent analysis. [2]
- **Ensure total protein content from each sample is standardized**
 - **So that differences subsequently detected in the differentially labeled fractions are due to intrinsic differences in levels of phosphoproteins between samples**
 - **Allows for accurate relative quantitation between wild-type and mutant samples**
- (b) Describe how the results of this investigation could provide useful information about the role of Yes kinase in the developmental events at the 24 hpf stage. [3]
- **MS-MS allows identification of proteins by peptide sequencing**
 - **Phosphopeptides that are significantly downregulated in Yes mutant sample & are (derived from proteins) phosphorylated by Yes kinase**
 - **Phosphorylation of the proteins important in normal development of donkeyfish embryos at 24 hpf stage**



- **By correlation with abnormalities / mutant phenotype(s) of Yes mutant donkeyfish embryos, can deduce which tissues / cells / developmental processes these phosphorylated proteins are important for**

(c) Briefly describe how RNAi is used to suppress the expression of Yes kinase protein. [2]
Introduce **double-stranded RNA** into the cell corresponding to **part of the gene sequence** so as to allow **silencing of gene at the post transcriptional level/suppress translation through rapid destruction of any mRNA synthesized.**

(d) With reference to **Fig. 7.2**, comment on the effectiveness of the length of the dsRNA on the expression of Yes kinase. [2]

- **the longer the dsRNA, the greater the silencing effect**
- **400 nt dsRNA most effective in gene silencing, producing 85% of the silencing effect**
- **200 nt dsRNA intermediate, compared to 400 nt dsRNA less effective, only about 55%**
- **least effective 50 nt dsRNA, controls are all at baseline level**

(e) Suggest why the use of RNAi makes it a valuable research tool in both cell culture and the study of living organisms. [1]

Highly selective, used for **large-scale screens that systematically shut down each gene in the cell.**

QUESTION 8

(a) Describe “cooperativity”. [2]

- It is a property **exhibited by allosteric enzyme** whereby the **binding of one substrate molecule to one of the active sites**
- **Shifts the enzyme to the R state, increasing substrate affinity for the rest of the active sites**

With reference to **Fig. 8.1**,

(b)(i) Describe the curve exhibited by enzyme X. [2]

- **As substrate concentration increases from 0 – 10 mM, rate of reaction increases slowly from 0 to 10 mgmin⁻¹**
- **Further increase in substrate concentration to 15 mM, causes a sharp increase in the rate of reaction to 80 mgmin⁻¹**
- **Rate of reaction slowly increases from 80 to 100 mgmin⁻¹ as the substrate concentration increases from 15 to 20 mM**
- **Further increase in the substrate concentration to 25mM does not lead to an increase in the rate of reaction / V_{max} is 100 mgmin⁻¹**

(ii) Explain the differences in the graph obtained by enzyme X and enzyme Y. [4]

V_{max} for enzyme X is greater than that of enzyme Y because for the same concentration of enzyme

(c) Suggest two other factors that can increase the rate of reaction catalyzed by enzyme Y. [2]

Factor 1: temperature, Factor 2: pH

QUESTION 9

(a) Describe how ATP synthase couples proton flux to ATP synthesis. [3]

The movement of protons through the F_o portion of ATP synthase causes the rotation of the F_o portion and its central stalk, thus leading to the cycling of the 3 catalytic nucleotide binding sites on the F₁ portion through the O (open), L (loose) and T (tight) states. This binding change mechanism results in ADP + P_i entering the binding site of the subunit in the O state, and the subsequent conformational changes through the L state and T state giving rise to the formation of ATP which is released when the subunit next cycles through the O state.

(b) Explain how the addition of an uncoupler to a suspension of mitochondria would affect the action of ATP synthase. [3]

Uncoupler allows the formation of proton channels in a proton-impermeable membrane which would allow for the free movement of protons down their concentration gradient. Hence there is no buildup of proton gradient and rate of proton movement through the F_o portion of the ATP synthase is slowed. The F₁ portion is not driven to cycle through the O, L and T states and rate of ATP synthesis will decrease drastically.

(c) When a proton gradient is generated by coupled electron transfer or ATP-hydrolysis, mitochondria take up Ca²⁺ from the external medium. The Ca²⁺-uptake is catalysed by a Ca²⁺ carrier in the inner membrane which exchanges Ca²⁺ for H⁺. What main structural features of this Ca²⁺ carrier would be necessary for it to carry out its function? [4]

- **Transmembrane protein to transfer Ca²⁺ and H⁺ across the membrane**
- **Ability to undergo conformational change in order to alternate between Ca²⁺ and H⁺ binding states on different sides of the membrane**
- **Possess Ca²⁺ and H⁺ binding sites that bind to their respective ions with differing affinity**
- **ATP binding site for ATP hydrolysis for conformational change**



QUESTION 10

Discuss with named examples how the following interactions are essential to cellular function.

(i) protein-protein interactions

[8]

Eg. 1

- a. chaperones with other proteins
- b. chaperones identify and bind to unfolded / misfolded proteins
- c. to aid in correct and proper folding of the proteins
- d. it also serves to maintain the protein in the unfolded state
- e. until the protein reaches the correct organelle / is in correct environment
- f. eg. of chaperones

Eg. 2

- a. receptor tyrosine kinases and signaling proteins
- b. upon ligand binding, RTKs dimerise
- c. and undergo autophosphorylation
- d. which serve as docking sites for intracellular signaling molecules
- e. to bind to and undergo phosphorylation and activation
- f. eg. of RTKs

Eg. 3

- a. immunoglobins and antigens
- b. synthesized by B cells
- c. each bearing a different antigen-binding site
- d. recognizing specific antigens and binding to them
- e. they also bear different tail regions with different functional properties
- f. such as ability to activate complement system / bind to phagocytic cells / cross the placenta from mother to fetus

Eg. 4

- a. F_0F_1 ATP synthase – multisubunit protein
- b. F_0 component consists of 3 protein subunits a, b and c
- c. Where the interface of a and c subunits form the proton channel
- d. F_1 component is a water-soluble complex of 5 distinct polypeptides $\alpha_3\beta_3\gamma\delta\epsilon$
- e. Interaction of these polypeptides forms the knob that protrude from matrix side of inner mitochondrial membrane
- f. Responsible for catalyzing ATP synthesis

Other eg. include Ca^{2+} pump, multienzyme complexes, hemoglobin, peptide hormones and their receptors, Fc receptors

(ii) protein-nucleic acid interactions

[12]

Eg. 1

- a. histones with DNA;
- b. negatively-charged DNA; binds with
- c. positively-charged lysine in histone tail;
- d. tight winding of 146bp of DNA;
- e. around histone octamer;
- f. H2a, H2b, H3 and H4;
- g. Forming a nucleosome;
- h. Responsible for further condensation into chromosome;

Eg. 2

- a. activators with enhancers;;
- b. promote binding of general transcription factors;
- c. and hence RNA polymerase;
- d. to promoter;
- e. achieving promoter clearance;
- f. increasing gene expression;
- g. different activators present in different cell types;
- h. leading to differential expression patterns and cell differentiation;



@ if answered with respect to repressors

Eg. 3

- a. tRNA with amino-acyl tRNA synthetase;;
- b. amino-acyl tRNA synthetase has active site complementary to tRNA;
- c. also recognizes anticodon sequence on tRNA;
- d. it functions to attach the corresponding amino acid;
- e. to 3'-CCA of the tRNA;
- f. through formation of covalent bond;
- g. tRNA is thus activated, enabling it to participate in translation;

Eg. 4

- a. helicase, ssBP, gyrase, DNA polymerase, RNA primase and DNA ligase with DNA;;
- b. helicase serves to unwind double-stranded DNA;
- c. ssBP and gyrase maintains the DNA in unwound state;
- d. so that RNA primase can bind to origin of replication (s) and synthesize RNA primers;
- e. to initiate DNA replication;
- f. that is catalysed by DNA polymerase;
- g. DNA ligase catalyses formation of phosphodiester linkages;
- h. Between Okazaki fragments / in the lagging strand;

Other eg. include restriction enzymes with DNA, Dicer in RNAi, translation factors such as eIF-4E, DNA methyltransferase, reverse transcriptase